



Global Stock Market Analysis

Team Members

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- Mohamed Sobhi Ibrahim (Power BI)
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- Injy Makarem Abd elmenaem (Tableau)
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Project Overview

Objective

The objective of this project is to analyze global stock market performance using historical financial data from multiple international stock indices. The project transforms raw market data into interactive dashboards and analytical reports using Excel, Power BI, Tableau, and Python, enabling users to monitor market trends, compare countries and indices, evaluate trading activity, and support data-driven investment decisions through visualization and forecasting.



Dataset

The dataset contains historical stock market data from multiple global indices, including:

- Date
- Index Name
- Country
- Open, High, Low, Close Prices
- Trading Volume
- Daily Change (%)

Additional analytical features were generated, such as:

- Daily Return
- Price Range
- Volatility
- Market Status
- Closing Position
- Price Direction



Data Preparation

- Loaded the dataset using Pandas.
- Inspected the dataset structure.
- Verified the Date column.
- Prepared the data for feature engineering.

```
df.head()
```

	Date	Index_Name	Country	Open	High	Low	Close	Volume	Daily_Change_Percent	Day_Of_Week
0	2020-01-01	S&P 500	USA	15607.06	16277.38	15427.47	16057.78	36788921	2.89	Wednesday
1	2020-01-01	NASDAQ Composite	USA	7083.79	7343.64	6461.54	6641.87	22249687	-6.24	Wednesday
2	2020-01-01	Dow Jones	USA	26384.65	26601.25	25659.50	25941.06	4344769	-1.68	Wednesday
3	2020-01-01	FTSE 100	UK	8091.17	8182.45	7617.15	7774.58	44250411	-3.91	Wednesday
4	2020-01-01	Nikkei 225	Japan	1899.43	2044.17	1885.43	1924.21	22498555	1.30	Wednesday

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 13050 entries, 0 to 13049
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Date                13050 non-null  datetime64[ns]
1   Index_Name          13050 non-null  object
2   Country              13050 non-null  object
3   Open                13050 non-null  float64
4   High                13050 non-null  float64
5   Low                 13050 non-null  float64
6   Close               13050 non-null  float64
7   Volume              13050 non-null  int64
8   Daily_Change_Percent 13050 non-null  float64
9   Day_Of_Week          13050 non-null  object
dtypes: datetime64[ns](1), float64(5), int64(1), object(3)
memory usage: 1819.7+ KB
```

Date Feature Extraction



To improve analysis, the Date column was transformed into multiple time-based features:

- Year
- Quarter
- Month
- Day
- Week

```
df["Year"] = df["Date"].dt.year
```

```
df.head()
```

	Date	Index_Name	Country	Open	High	Low	Close	Volume	Daily_Change_Percent	Day_Of_Week	Year
0	2020-01-01	S&P 500	USA	15607.06	16277.38	15427.47	16057.78	36788921	2.89	Wednesday	2020
1	2020-01-01	NASDAQ Composite	USA	7083.79	7343.64	6461.54	6641.87	22249687	-6.24	Wednesday	2020
2	2020-01-01	Dow Jones	USA	26384.65	26601.25	25659.50	25941.06	4344769	-1.68	Wednesday	2020
3	2020-01-01	FTSE 100	UK	8091.17	8182.45	7617.15	7774.58	44250411	-3.91	Wednesday	2020
4	2020-01-01	Nikkei 225	Japan	1899.43	2044.17	1885.43	1924.21	22498555	1.30	Wednesday	2020

```
df["Quarter"] = df["Date"].dt.quarter
```

```
df.head()
```

	Date	Index_Name	Country	Open	High	Low	Close	Volume	Daily_Change_Percent	Day_Of_Week	Year	Quarter
0	2020-01-01	S&P 500	USA	15607.06	16277.38	15427.47	16057.78	36788921	2.89	Wednesday	2020	1
1	2020-01-01	NASDAQ Composite	USA	7083.79	7343.64	6461.54	6641.87	22249687	-6.24	Wednesday	2020	1
2	2020-01-01	Dow Jones	USA	26384.65	26601.25	25659.50	25941.06	4344769	-1.68	Wednesday	2020	1
3	2020-01-01	FTSE 100	UK	8091.17	8182.45	7617.15	7774.58	44250411	-3.91	Wednesday	2020	1

These features help analyze seasonal trends and improve forecasting accuracy.

Feature Engineering



Creating Analytical Features

Several new features were generated to enhance stock market analysis:

- Price Range = High – Low
- Daily Profit = Close – Open
- Daily Return (%)
- Volatility (%)
- Market Status (Bullish / Bearish / Neutral)
- Closing Position
- Price Direction
- Volatility Level

```
df["Price_Range"] = df["High"] - df["Low"]
```



```
df.head()
```

	Date	Index_Name	Country	Open	High	Low	Close	Volume	Daily_Change_Percent	Day_Of_Week	Year	Quarter	Month	
0	2020-01-01	S&P 500	USA	15607.06	16277.38	15427.47	16057.78	36788921	2.89	Wednesday	2020	1	January	Wedne
1	2020-01-01	NASDAQ Composite	USA	7083.79	7343.64	6461.54	6641.87	22249687	-6.24	Wednesday	2020	1	January	Wedne
2	2020-01-01	Dow Jones	USA	26384.65	26601.25	25659.50	25941.06	4344769	-1.68	Wednesday	2020	1	January	Wedne
3	2020-01-01	FTSE 100	UK	8091.17	8182.45	7617.15	7774.58	44250411	-3.91	Wednesday	2020	1	January	Wedne
4	2020-01-01	Nikkei 225	Japan	1899.43	2044.17	1885.43	1924.21	22498555	1.30	Wednesday	2020	1	January	Wedn

These engineered features provide better insights for visualization and forecasting models.

Power BI Dashboard Overview



- Interactive dashboard for global stock market analysis.
- Built using DAX Measures.
- Dynamic filtering by Country, Year, Month and Index.
- Provides real-time analytical insights.



Power BI Dashboard Overview



Power BI Dashboard Overview



Power BI Dashboard Dashboard Insights

KPIs

- Total Markets
- Average Close
- Maximum Close
- Minimum Close
- Average Return
- Total Volume

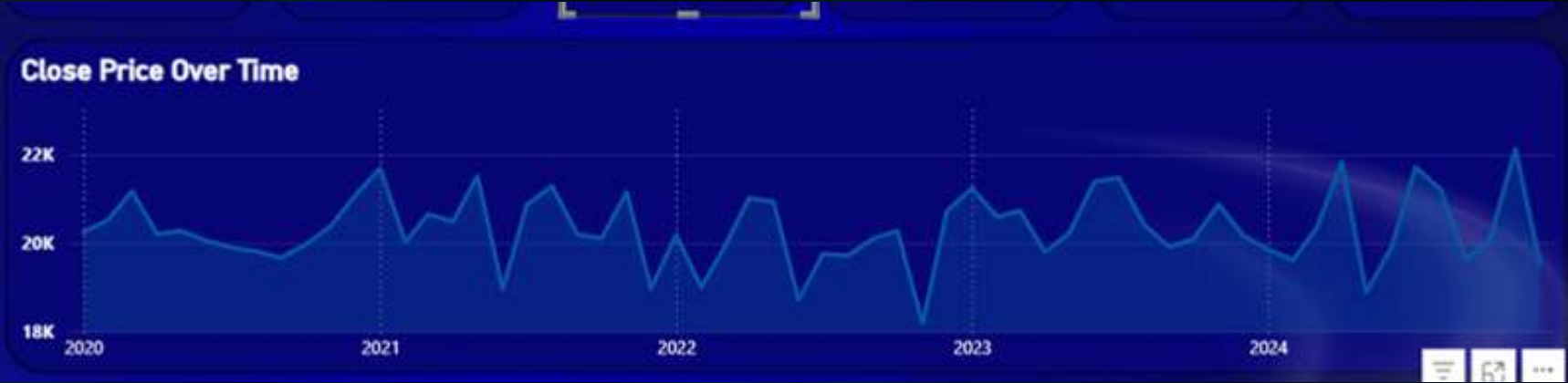


Power BI Dashboard Dashboard Insights

Visualizations

- Area Chart
- Azure Map
- Treemap
- Donut Chart
- Column Chart

Country	Average Close	Average Return	Total Volume	Average Volatility
UK	19,949.98	-0.01	32736235707	5.51
Japan	20,195.94	-0.21	32895021824	5.39
Pakistan	20,504.42	0.13	33142417272	5.25
France	20,394.17	-0.08	33471252301	5.08
China	20,171.76	-0.03	33604880170	5.16
Germany	20,278.87	0.01	33795280766	5.03
Hong Kong	20,670.66	0.08	33940276909	5.24
USA	20,352.97	-0.11	100052608789	5.21
Total	20,322.47	-0.04	333637973738	5.23



Excel Dashboard Overview



KPIs & Charts

- Close Price Trend
- Index Performance Comparison
- Revenue & Profit Analysis
- Index Performance Ranking
- Interactive Pivot Tables & Slicers



Excel Dashboard Overview



- Interactive dashboard built using Pivot Tables and Pivot Charts.
- Analyze stock market performance across countries and indices.
- Dynamic filtering using Slicers.
- Supports quick comparison of financial metrics.



Tableau Dashboard Overview

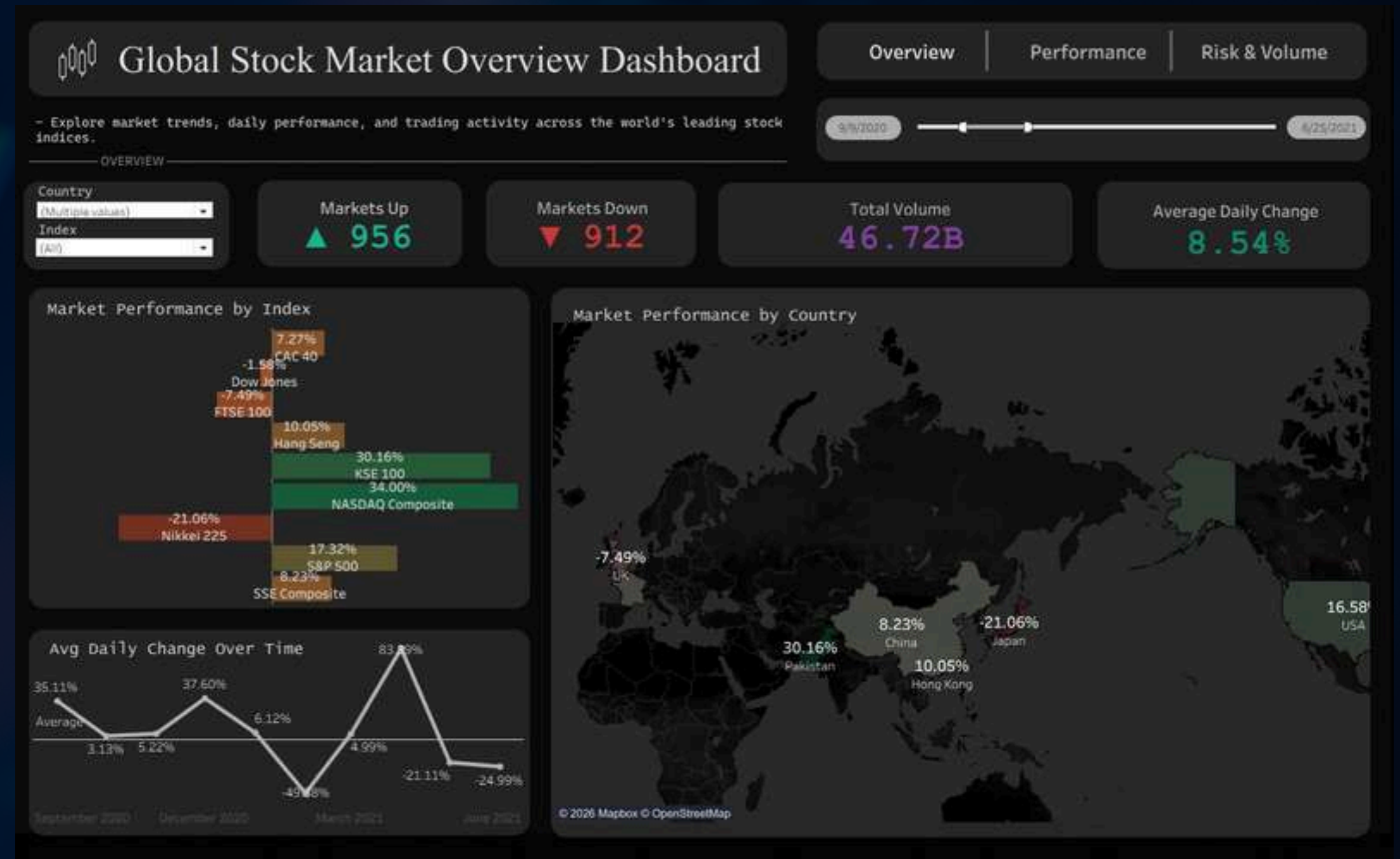


Three interactive dashboards were developed:

- Global Market Overview
- Performance Analysis
- Trading Activity & Risk

Key Visualizations

- Filled Map
- Candlestick Chart
- Risk vs Return Scatter Plot
- Trading Volume Analysis
- KPI Cards
- Bubble Charts



They provide market performance, trading activity and risk-return analysis.

Tableau Dashboard Overview

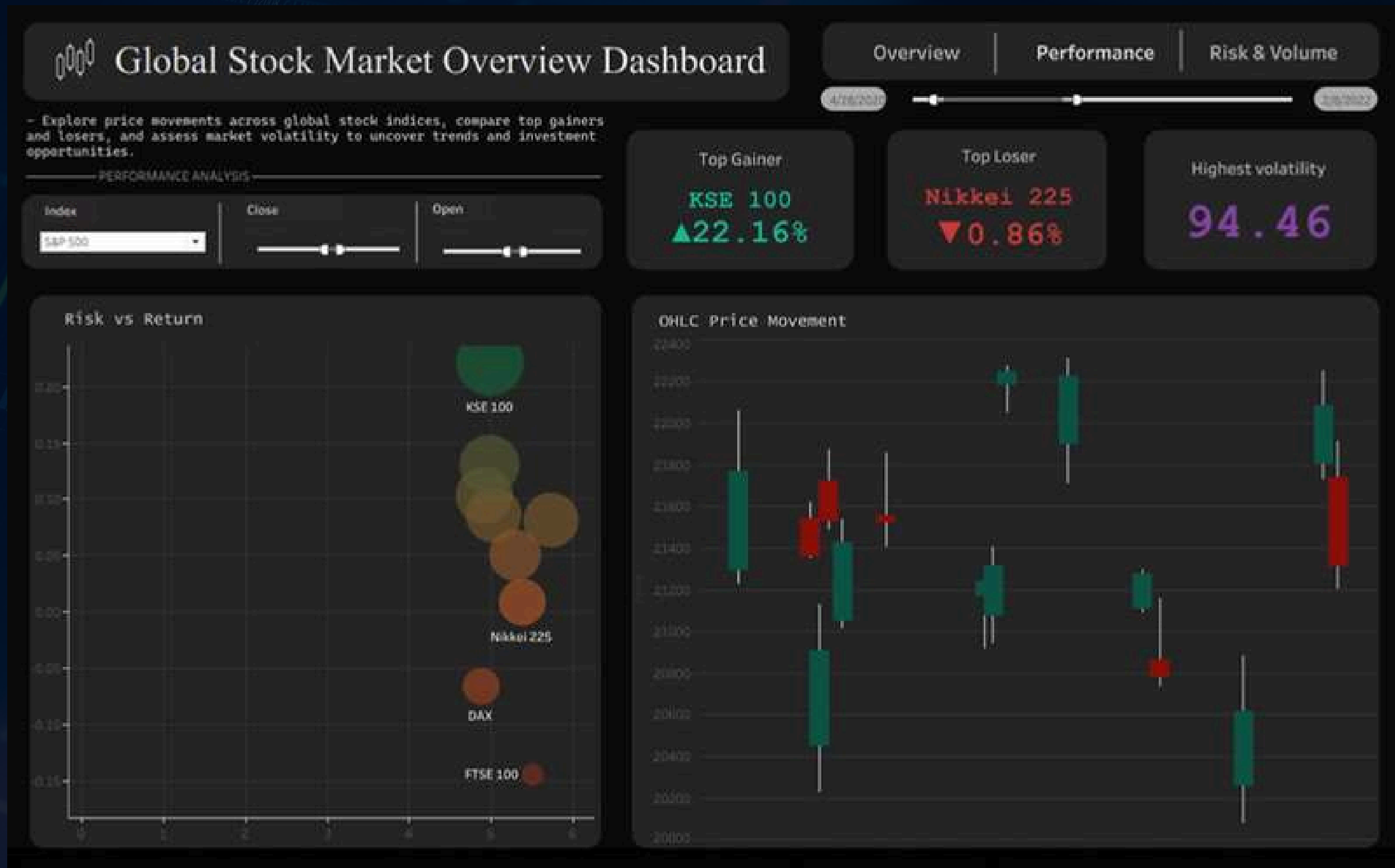
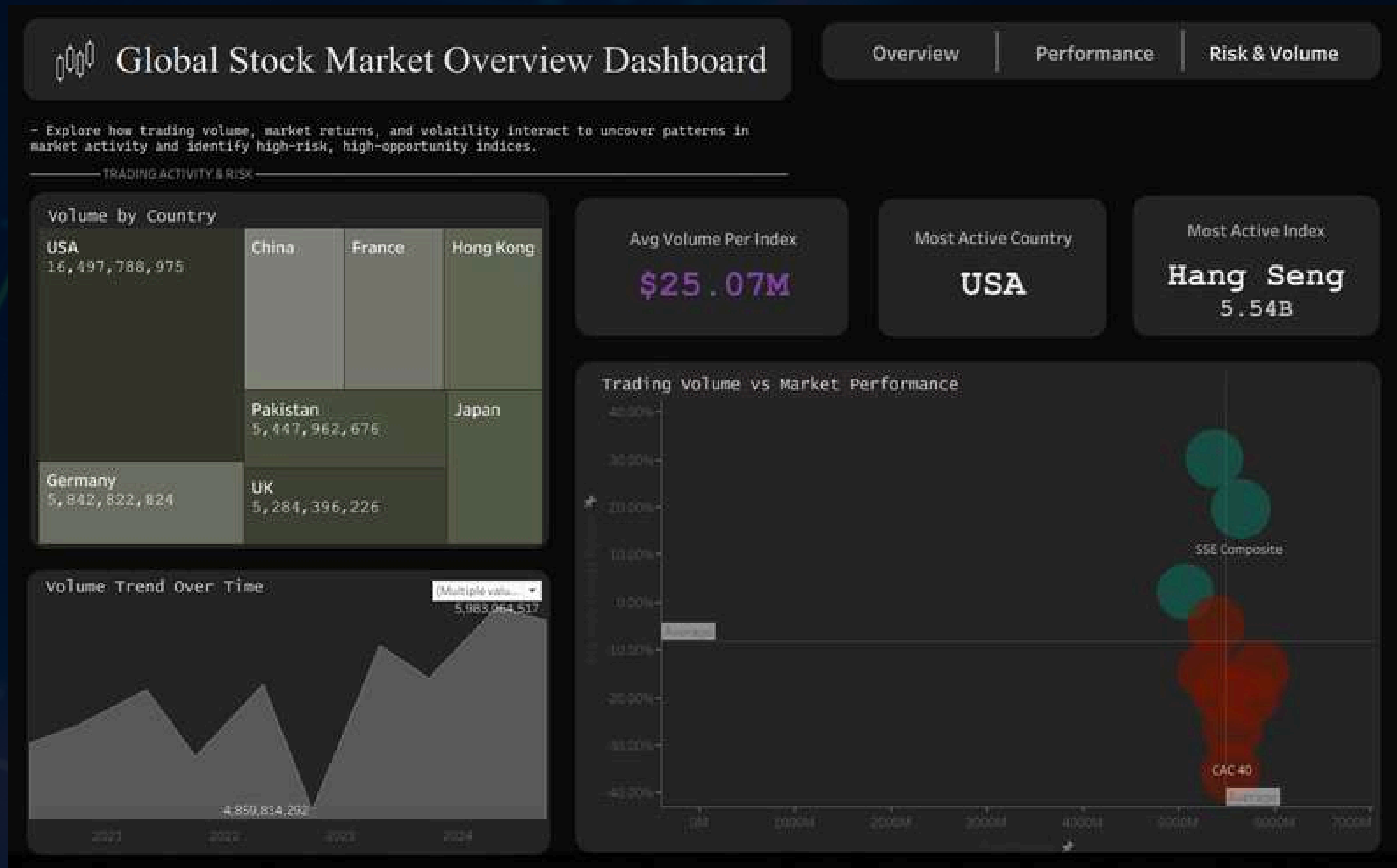


Tableau Dashboard Overview



Python Dashboard Overview

Provides a high-level summary of the global stock market using key performance indicators and interactive visualizations.

Key Metrics (KPIs)

- **Total Volume:** Total trading volume across selected markets.
- **Average Daily Change:** Average daily percentage change.
- **Top Gainer:** Index with the highest average daily return.
- **Top Loser:** Index with the lowest average daily return.

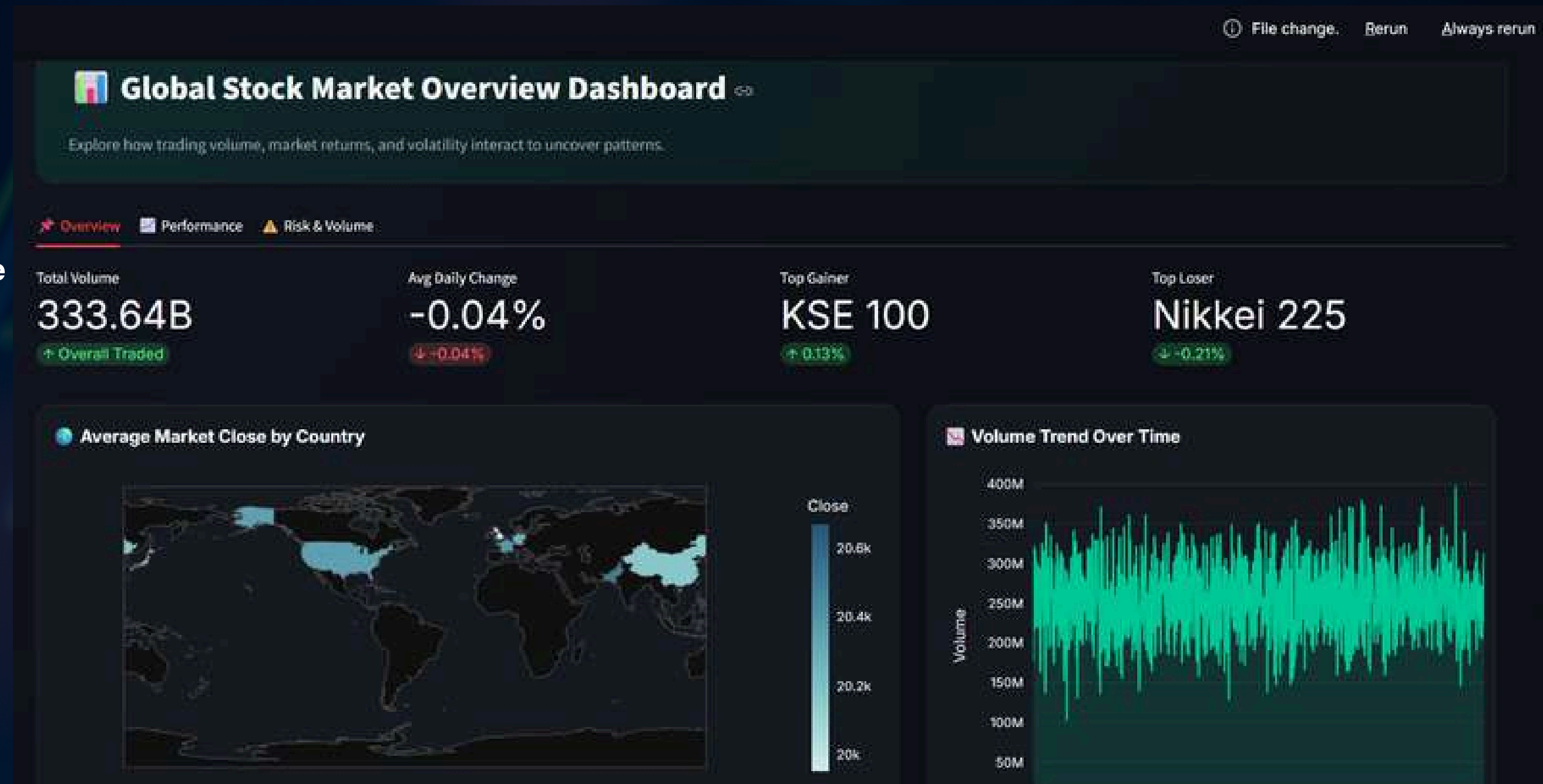
Visualizations

Choropleth Map – Average Market Close by Country

Displays the average closing price for each country.
Helps compare market performance across different regions.

Area Chart – Volume Trend Over Time

Shows how trading volume changes over time.
Identifies periods of high and low market activity.



Python Dashboard Overview

Performance Analysis

Focuses on market performance, price movement, and return comparison across countries.

Visualizations

Treemap – Volume by Country & Index

- Displays trading volume grouped by country and stock index.
- Larger rectangles represent higher trading volumes.
- Helps identify the most active markets.

Candlestick Chart – OHLC Price Movement

- Visualizes daily price movement using:
- Green candles indicate price increases.
- Red candles indicate price decreases.
- Commonly used for technical analysis and trend identification.

Bar Chart – Average Daily Return by Country

- Compares the average daily return of each country.
- Bars above zero indicate positive returns.
- Bars below zero indicate negative returns.
- Makes it easy to compare market performance.



Python Dashboard Overview

Risk Analysis

Analyzes the relationship between market risk, trading activity, and investment returns.

Visualizations



Bubble Chart – Risk vs Return

- Evaluates the balance between risk and return.
- Identifies indices with strong performance relative to their risk level.



Bubble Chart – Trading Volume vs Market Performance

- Examines the relationship between trading activity and market performance.
- Shows whether higher trading volumes are associated with higher or lower returns.



The background is a dark blue, textured image featuring a faint world map. Overlaid on the map are various financial data visualizations, including candlestick charts, line graphs, and numerical data points. The overall aesthetic is professional and tech-oriented, typical of a financial or trading interface.

**Thank
You**